# Self-naturality of a homotopy from f to itself: H(f(x)) equals ap of f applied to H(x)

The Univalent Foundations Program

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*Editorial note.* The counted statement is the author corollary labelled cor:hom-fg (source lines 1046-1049); the proof span 1052-1068 is the author's proof environment, retained line for line.

*Editorial note.* The author proof suppresses some associativity re-association paths (its own closing remark); these are instances of the associativity law for concatenation, not missing cases, and the card retains the author's wording.

*Editorial note.* The naturalness invoked by the proof is supplied by the retained context.

*Editorial scope.* One original result of the HoTT book chapter cha:basics, counted once: the self-naturality identification for a homotopy of an endofunction. Only the counted corollary and its author proof are retained; the author's commentary between statement and proof is background.

*Allowed standard prerequisites (editorial).*

- Homotopies between endofunctions of A and the action of a function on paths (ap), with its definitional behaviour on reflexivity.
- Path concatenation and inversion with the unit laws and associativity of concatenation.
- Naturality of homotopies (the retained context).

**Lemma 0.0.1.** *Suppose $H : f \sim g$ is a homotopy between functions $f, g : A \to B$ and let $p : x =_A y$. Then we have*

$$H(x) \cdot g(p) = f(p) \cdot H(y).$$

*We may also draw this as a commutative diagram:*

$$\begin{array}{ccc} f(x) & \xlongequal{f(p)} & f(y) \\ {\scriptstyle H(x)}\Big\| & & \Big\|{\scriptstyle H(y)} \\ g(x) & \xlongequal[g(p)]{} & g(y) \end{array}$$

**Corollary 0.0.2.** *Let $H : f \sim \mathsf{id}_A$ be a homotopy, with $f : A \to A$. Then for any $x : A$ we have*

$$H(f(x)) = f(H(x)).$$

*Proof.* By naturality of $H$, the following diagram of paths commutes:

$$\begin{array}{ccc} ffx & \xlongequal{f(Hx)} & fx \\ {\scriptstyle H(fx)}\Big\| & & \Big\|{\scriptstyle Hx} \\ fx & \xlongequal[Hx]{} & x \end{array}$$

That is, $f(Hx) \cdot Hx = H(fx) \cdot Hx$. We can now whisker by $(Hx)^{-1}$ to cancel $Hx$, obtaining

$$f(Hx) = f(Hx) \cdot Hx \cdot (Hx)^{-1} = H(fx) \cdot Hx \cdot (Hx)^{-1} = H(fx)$$

as desired (with some associativity paths suppressed). □